ELLA BRECH

Computer Science & Cybersecurity Student | Software & Security Enthusiast

(920) 745-0197 (please leave a voicemail if I don't pick up) | portfolio.ellasbox.net

SUMMARY

Computer Science student and systems-minded developer who builds reliable software and infrastructure, from backend scripting to self-hosted Linux services to hardened networks. Active cybersecurity researcher and Collegiate Cyber Defense Competition team member with a track record of shipping open-source tooling and squeezing the most out of anything she is given. Seeking software development or IT/security roles that reward curiosity and technical precision.

TECHNICAL SKILLS

Software Development: Scripting, web development, CI/CD, databases, Git, open-source contribution

Systems & DevOps: Linux (Debian, RHEL, Arch), Docker, SystemD, Apache, reverse proxies, Tailscale

Networking & Security: TCP/IP, routing, SD-WAN, network/security design, malware analysis & threat research

AV & Systems Integration: Q-SYS, Dante/AVoIP, live A/V system design, lighting design (ETC Eos/Hog)

SELECTED PROJECTS

Homelab (Sysadmin): Running a self-hosted multi-host Linux environment since 2025 with Dockerized services, an Apache reverse proxy, and Tailscale for secure remote access with access control.

Collegiate Cyber Defense Competition: Competing on UW-Stout Polytechnic's 2026–27 team, hardening live systems and services against an active red team.

Team 5 Consulting: Designed and proposed network solutions (hardware, SD-WAN, services, security) across client scales for a Network Systems & Design course project.

EXPERIENCE

Cybersecurity Researcher — DeTraced Security *Jun 2026 – Present | Remote*

- Design, develop, and document security tools (e.g., YARlint), collaborating with a team to contribute to and maintain open-source software repositories.
- Research emerging cybersecurity threats, including malware analysis, and support takedown efforts against malicious infrastructure and threat actors.
- Participate in cybersecurity competitions on behalf of DeTraced, competing against leading groups in the space.

Audio/Visual Technician — UW-Stout Event Tech Crew *Feb 2025 – Present | Menomonie, WI*

- Design and deploy A/V, network, and lighting solutions for spaces ranging from small classrooms to large event venues.
- Lead live event audio operations, applying quick decision-making and technical precision under live conditions.

Communications Director, Executive Board — Blue Devil Productions *Nov 2024 – Present | Menomonie, WI*

- Manage event data and public engagement information within event management software; built a campus marketing distribution system for prior and prospective attendees.
- Coordinate communications between organization staff and the public to plan and execute campus events.

Customer Service Associate & Projectionist — Army & Air Force Exchange Svcs *Dec 2022 – Aug 2023 | Baumholder, Germany*

- Operated and maintained digital theater projection and audio playback systems alongside customer service and ticketing duties.

EDUCATION

University of Wisconsin-Stout Polytechnic *2024 – 2028*

B.S., Computer Science with a concentration in Cybersecurity & Secure Software Development

CERTIFICATIONS

TestOut Linux Pro · Dante Level 3 + AV · Q-SYS Level 0 · Adobe Certified Professional, Digital Video (Premiere Pro)